

SHELAGONTSY, Russian Federation

WMO: 243290

Lat: 66.250N

Lon: 114.283E

Elev: 236

StdP: 98.52

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -54.1	(c) -51.9	(d) -55.1	(e) 0.0	(f) -54.0	(g) -53.0	(h) 0.0	(i) -51.7	(j) 7.1	(k) -19.4	(l) 5.6	(m) -23.2	(n) 0.3	(o) 230	(p) 0.772

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.7	(c) 28.4	(d) 17.4	(e) 25.9	(f) 16.3	(g) 23.4	(h) 15.0	(i) 18.5	(j) 25.5	(k) 17.3	(l) 23.9	(m) 16.0	(n) 22.1	(o) 1.9	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.7	20.9	14.6	10.7	19.8	13.2	9.8	18.4	53.6	25.4	49.6	24.1	45.9	22.2	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.5	(b) 5.3	(c) 4.5	(e) -54.2	(f) 31.5	(g) 2.1	(h) 1.9	(i) -55.7	(j) 32.9	(k) -56.9	(l) 34.0	(m) -58.1	(n) 35.0	(o) -59.6	(p) 36.3		
(4) 6.5	(5) 5.3	(6) 4.5	(7) -54.1	(8) 21.2	(9) 2.0	(10) 1.3	(11) -55.6	(12) 22.1	(13) -56.8	(14) 22.9	(15) -57.9	(16) 23.6	(17) -59.4	(18) 24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.7	-38.7	-35.0	-23.2	-9.6	1.6	12.0	14.8	10.6	2.3	-10.6	-28.6	-38.1
	DBStd	20.65	9.17	8.42	8.74	6.96	5.63	4.69	3.98	3.73	4.48	8.15	8.83	8.13
	HDD10.0	8246	1511	1259	1028	588	264	31	6	37	232	639	1158	1491
	HDD18.3	10999	1769	1492	1287	838	519	196	122	240	480	897	1408	1749
	CDD10.0	307	0	0	0	0	4	91	155	56	2	0	0	0
	CDD18.3	18	0	0	0	0	0	5	12	1	0	0	0	0
	CDH23.3	512	0	0	0	0	2	141	322	47	1	0	0	0
	CDH26.7	139	0	0	0	0	0	34	95	10	0	0	0	0
(14) Wind	WSAvg	1.2	0.7	0.8	1.2	1.6	1.9	1.6	1.4	1.3	1.3	1.2	0.9	0.6
(15) Precipitation	PrecAvg	292	9	7	9	13	27	45	57	46	29	22	15	11
	PrecMax	443	23	16	23	43	50	99	172	116	55	39	32	26
	PrecMin	174	2	1	2	1	2	6	1	8	4	9	5	0
	PrecStd	66	4	4	5	8	13	24	38	27	15	8	6	5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.3	-7.8	2.9	11.2	22.7	30.6	31.8	27.9	18.8	6.2	-2.0	-12.4
		MCWB	-9.1	-8.6	-0.4	4.5	11.4	17.4	19.4	17.8	11.5	3.7	-2.8	-13.1
	2%	DB	-13.9	-12.7	-1.2	7.6	18.0	27.4	29.3	23.9	14.2	3.7	-6.0	-16.6
		MCWB	-14.4	-13.3	-3.7	2.4	9.4	16.6	17.9	15.7	9.4	2.2	-6.8	-17.1
	5%	DB	-19.0	-17.4	-4.6	4.6	14.5	24.9	27.0	21.6	11.8	1.7	-11.0	-21.7
		MCWB	-19.3	-17.8	-6.3	0.8	7.5	15.2	17.0	14.4	7.6	0.5	-11.6	-21.9
	10%	DB	-25.0	-21.7	-7.9	2.4	11.4	22.3	24.7	19.1	9.8	-0.1	-14.8	-25.3
		MCWB	-25.1	-21.9	-9.3	-0.6	5.7	13.8	16.2	13.3	6.6	-1.3	-15.2	-25.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.9	-8.6	-0.3	4.6	12.2	19.2	20.7	18.4	12.8	4.0	-2.6	-12.9
		MCDB	-8.4	-7.7	2.8	10.5	21.2	26.1	28.2	25.1	16.3	5.8	-2.1	-12.4
	2%	WB	-14.5	-13.3	-3.6	2.8	9.8	17.5	19.2	16.7	9.9	2.4	-6.8	-17.1
		MCDB	-14.0	-12.7	-1.4	6.8	16.6	25.1	26.4	22.2	12.5	3.6	-6.0	-16.6
	5%	WB	-19.4	-17.8	-6.2	1.2	7.8	16.1	18.0	15.3	8.4	0.5	-11.5	-21.9
		MCDB	-19.1	-17.3	-4.5	4.3	13.9	23.5	24.7	19.9	11.3	1.6	-11.0	-21.7
	10%	WB	-25.2	-22.0	-9.3	-0.5	6.0	14.6	16.9	13.9	6.9	-1.2	-15.1	-25.5
		MCDB	-25.1	-21.6	-8.0	2.3	11.1	21.1	23.3	18.3	9.3	-0.2	-14.7	-25.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	12.6	18.9	17.1	11.5	12.5	13.7	13.1	9.9	9.3	10.0	8.8
		MCDBR	14.1	15.6	18.1	17.7	16.1	17.9	19.0	18.8	14.7	9.2	13.5	14.0
		MCWBR	13.6	15.3	16.0	13.4	9.3	8.7	9.0	10.9	9.8	7.1	12.9	13.8
	5% WB	MCDBR	13.9	15.6	17.7	15.8	15.1	15.7	16.0	15.4	12.9	8.0	13.2	13.9
		MCWBR	13.6	15.3	15.7	12.2	9.3	8.3	8.8	9.8	9.1	6.7	12.8	13.6
(40) Clear-Sky Solar Irradiance	taub		0.186	0.237	0.285	0.314	0.344	0.373	0.415	0.419	0.308	0.249	0.178	0.179
	taud		1.871	2.057	2.075	2.103	2.254	2.387	2.257	2.172	2.482	2.205	1.893	1.879
	Ebn at Noon		225	619	762	838	852	835	776	719	765	604	222	25
	Edh at Noon		18	55	92	117	113	103	114	112	64	47	15	4
(44) All-Sky Solar Radiation	RadAvg		0.10	0.84	2.56	4.63	5.57	5.48	5.14	3.48	1.98	0.85	0.21	0.01
	RadStd		0.01	0.04	0.11	0.23	0.30	0.47	0.34	0.42	0.20	0.11	0.01	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page